

Lab Sheet 8: ES6 Basics

Branch/ Class: B.Tech/M.Tech

Date:

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name:

Reg. no.:

Exercise 1

Write a JavaScript program using ES6 features to calculate the total and average marks of a student. The program should declare variables using `let` and `const`, use an arrow function to compute the average, and display the result using template literals. The program should print the student name, total marks, and average marks in a formatted message.

Sample Code Structure

```
let studentName = "Arun";
let mark1 = 85;
let mark2 = 90;
let mark3 = 88;
const calculateAverage = (m1, m2, m3) => {
  return (m1 + m2 + m3) / 3;
}
let average = calculateAverage(mark1, mark2, mark3);
console.log(`Student Name: ${studentName}`);
console.log(`Average Marks: ${average}`);
```

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```
JS Ex1.js  X  JS Ex2.js  JS Ex3.js
Lab08 > JS Ex1.js > ...
1  let studentName = "Arun";
2  let mark1 = 85;
3  let mark2 = 90;
4  let mark3 = 88;
5
6
7  let total = mark1 + mark2 + mark3;
8
9
10 const calculateAverage = (m1, m2, m3) => {
11   |   return ((m1 + m2 + m3) / 3).toFixed(2);
12 };
13
14 let average = calculateAverage(mark1, mark2, mark3);
15
16 console.log(`Student Name: ${studentName}`);
17 console.log(`Total Marks: ${total}`);
18 console.log(`Average Marks: ${average}`);
```

Expected Output

Student Name: Arun

Average Marks: 87.66

```
PS C:\Users\shalemraju nakkala\OneDrive\Desktop\webtech\Lab08> node Ex1.js
Student Name: Arun
Total Marks: 263
Average Marks: 87.67
```

Exercise 2

Create a JavaScript program that defines an object representing a student with properties such as id, name, department, and marks. Use object destructuring to extract the values and display them. Then create a new object using the spread operator that adds an additional property called grade based on the marks and display the updated object.

Sample Code Structure

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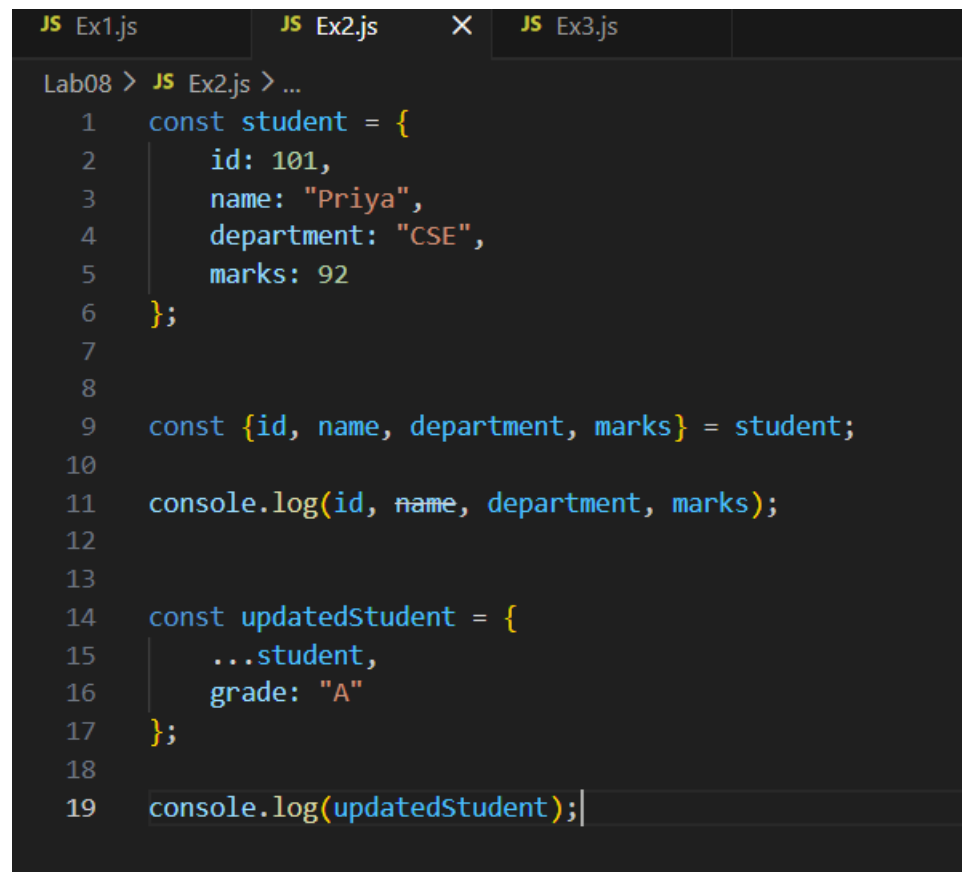
School: SCOPE

Student name:

Reg. no.:

```
const student = {  
  id: 101,  
  name: "Priya",  
  department: "CSE",  
  marks: 92  
};
```

```
const {id, name, department, marks} = student;  
console.log(id, name, department, marks);  
const updatedStudent = {  
  ...student,  
  grade: "A"  
};  
console.log(updatedStudent);
```



The screenshot shows a code editor with three tabs: 'JS Ex1.js', 'JS Ex2.js' (active), and 'JS Ex3.js'. The active tab contains the following code:

```
Lab08 > JS Ex2.js > ...  
1  const student = {  
2    id: 101,  
3    name: "Priya",  
4    department: "CSE",  
5    marks: 92  
6  };  
7  
8  
9  const {id, name, department, marks} = student;  
10  
11 console.log(id, name, department, marks);  
12  
13  
14 const updatedStudent = {  
15   ...student,  
16   grade: "A"  
17 };  
18  
19 console.log(updatedStudent);
```

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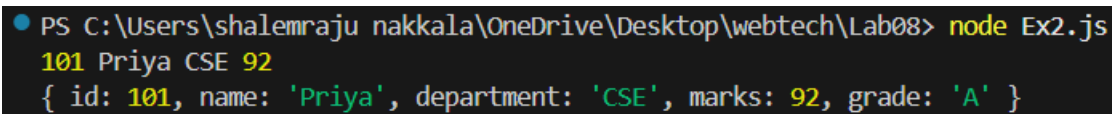
Student name:

Reg. no.:

Expected Output

101 Priya CSE 92

{ id: 101, name: 'Priya', department: 'CSE', marks: 92, grade: 'A' }



```
PS C:\Users\shalemraju nakkala\OneDrive\Desktop\webtech\Lab08> node Ex2.js
101 Priya CSE 92
{ id: 101, name: 'Priya', department: 'CSE', marks: 92, grade: 'A' }
```

Exercise 3

Develop a JavaScript program using ES6 classes and promises to simulate a simple course enrollment system. Create a class named Course with a constructor to initialize the course name and instructor. Include a method to display course details. Implement a promise that checks whether seats are available and prints either "Enrollment Successful" or "Course Full" based on the condition.

Sample Code Structure

```
class Course {
  constructor(courseName, instructor) {
    this.courseName = courseName;
    this.instructor = instructor;
  }
  displayCourse() {
    console.log(`Course: ${this.courseName}, Instructor: ${this.instructor}`);
  }
}

let course1 = new Course("Web Technologies", "Dr. Kumar");
course1.displayCourse();
let enrollCourse = new Promise((resolve, reject) => {
  let seatsAvailable = true;
  if(seatsAvailable)
    resolve("Enrollment Successful");
  else
    reject("Course Full");
});
enrollCourse
.then(msg => console.log(msg))
```

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.catch(err => console.log(err));

```
JS Ex1.js JS Ex2.js JS Ex3.js X
Lab08 > JS Ex3.js > Course > displayCourse
1  class Course {
2      constructor(courseName, instructor) {
3          this.courseName = courseName;
4          this.instructor = instructor;
5      }
6
7      displayCourse() {
8          console.log(`Course: ${this.courseName}, Instructor: ${this.instructor}`);
9      }
10 }
11
12
13 let course1 = new Course("Web Technologies", "Dr. Kumar");
14
15
16 course1.displayCourse();
17
18
19 let enrollCourse = new Promise((resolve, reject) => {
20
21     let seatsAvailable = true;
22
23     if(seatsAvailable)
24         resolve("Enrollment Successful");
25     else
26         reject("Course Full");
27 });
28
29
30
31 enrollCourse
32 .then(msg => console.log(msg))
33 .catch(err => console.log(err));
```

Expected Output

Course: Web Technologies, Instructor: Dr. Kumar

Enrollment Successful

VIT-AP UNIVERSITY, ANDHRA PRADESH

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```
PS C:\Users\shalemraju nakkala\OneDrive\Desktop\webtech\Lab08> node Ex3.js
Course: Web Technologies, Instructor: Dr. Kumar
Enrollment Successful
```